Lecture 3: Boundary Problems

3.1 Fourier Series

In this section, we will introduce how to find the coefficients in a Fourier series, which is very important in this lecture note. Let us first consider the full Fourier series for $x \in (-l, l)$]:

$$\boxed{\phi(x) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty}\left(A_n \cos\frac{n\pi x}{l} + B_n \sin\frac{n\pi x}{l}\right).} \tag{3.1}$$

Theorem 3.1 *The coeffiecients A_n and B_n can be represented in the following formulas:*

$$\boxed{A_n = \frac{1}{l}\int_{-l}^{l}\phi(x)\cos\frac{n\pi x}{l}dx \qquad (n = 0, 1, 2, ...) \quad \textit{and} \quad B_n = \frac{1}{l}\int_{-l}^{l}\phi(x)\sin\frac{n\pi x}{l}dx \qquad (n = 1, 2, 3, ...).} \tag{3.2}$$

Proof: The proof is based on the following observations.

$$\begin{aligned}
\int_{-l}^{l}\cos\frac{n\pi x}{l}\sin\frac{m\pi x}{l}dx &= 0, && \text{for all } n, m\\
\int_{-l}^{l}\cos\frac{n\pi x}{l}\cos\frac{m\pi x}{l}dx &= 0, && \text{for } n \neq m\\
\int_{-l}^{l}\sin\frac{n\pi x}{l}\sin\frac{m\pi x}{l}dx &= 0, && \text{for } n \neq m\\
\int_{-l}^{l}\cos\frac{n\pi x}{l}dx = \int_{-l}^{l}\sin\frac{n\pi x}{l}dx &= 0, && \text{for all } n\\
\int_{-l}^{l}\cos^2\frac{n\pi x}{l}dx = \int_{-l}^{l}\sin^2\frac{n\pi x}{l}dx &= l, && \text{for all } n
\end{aligned}$$

Let us fix m, multiply (3.1) by $\sin\frac{m\pi x}{l}$ and integrate the series (3.1) term by term to get

$$\int_{-l}^{l}\phi(x)\sin\frac{m\pi x}{l}dx = B_m\int_{-l}^{l}\sin^2\frac{m\pi x}{l}dx = B_m l.$$

The same to multiply $\cos\frac{m\pi x}{l}$ to get the formula of A_m. ■

Next, let us consider the Fourier sine series in the interval $(0, l)$.

Theorem 3.2 *The coefficients A_n in the Fourier sine series*

$$\boxed{\phi(x) = \sum_{n=1}^{\infty}A_n \sin\frac{n\pi x}{l}} \tag{3.3}$$

in the interval $(0, l)$ *are*

$$\boxed{A_n = \frac{2}{l}\int_0^l \phi(x)\sin\frac{n\pi x}{l}dx.} \tag{3.4}$$

We omit the proof here since it is similar to the precedent one. Meanwhile, we can also have the following theorem which corresponds to the cosine series.

Theorem 3.3 *The coefficients* A_n *in the Fourier cosine series*

$$\boxed{\phi(x) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \cos\frac{n\pi x}{l}} \tag{3.5}$$

in the interval $(0, l)$ *are*

$$\boxed{A_n = \frac{2}{l}\int_0^l \phi(x)\cos\frac{n\pi x}{l}dx.} \tag{3.6}$$

Example 1.

Let $\phi(x) = x$ in the interval $(0, l)$. Its Fourier sine series has the oefficients

$$A_n = \frac{2}{l}\int_0^l x\sin\frac{n\pi x}{l}dx = -\frac{2x}{n\pi}\cos\frac{n\pi x}{l} + \frac{2l}{n^2\pi^2}\sin\frac{n\pi x}{l}\Big|_0^l = (-1)^{n+1}\frac{2l}{n\pi}.$$

Thus in $(0, l)$ we have

$$x = \frac{2l}{\pi}\left(\sin\frac{\pi x}{l} - \frac{1}{2}\sin\frac{2\pi x}{l} + \frac{1}{3}\sin\frac{3\pi x}{l} - \cdots\right).$$

Example 2.

Let $\phi(x) = x$ in the interval $(0, l)$. Its Fourier cosine series has the oefficients

$$A_0 = \frac{2}{l}\int_0^l x dx = l,$$

$$A_n = \frac{2}{l}\int_0^l x\cos\frac{n\pi x}{l}dx = \frac{2x}{n\pi}\sin\frac{n\pi x}{l} + \frac{2l}{n^2\pi^2}\cos\frac{n\pi x}{l}\Big|_0^l = -\frac{4l}{n^2\pi^2}$$

for n odd, and 0 for n even. Thus in $(0, l)$ we have

$$x = \frac{l}{2} - \frac{4l}{\pi^2}\left(\cos\frac{\pi x}{l} + \frac{1}{9}\cos\frac{3\pi x}{l} + \frac{1}{25}\cos\frac{5\pi x}{l} + \cdots\right).$$

Example 3.

Let $\phi(x) = x$ in the interval $(-l, l)$. Its full Fourier series has the oefficients

$$A_0 = \frac{1}{l}\int_{-l}^l x dx = 0,$$

$$A_n = \frac{1}{l}\int_{-l}^l x\cos\frac{n\pi x}{l}dx = \frac{x}{n\pi}\sin\frac{n\pi x}{l} + \frac{l}{n^2\pi^2}\cos\frac{n\pi x}{l}\Big|_{-l}^l = 0$$

$$B_n = \frac{1}{l}\int_{-l}^{l} x \sin\frac{n\pi x}{l}dx = -\frac{x}{n\pi}\cos\frac{n\pi x}{l} + \frac{l}{n^2\pi^2}\sin\frac{n\pi x}{l}\Big|_{-l}^{l} = (-1)^{n+1}\frac{2l}{n\pi}.$$

Thus in $(0, l)$ we have

$$x = \frac{2l}{\pi}\left(\sin\frac{\pi x}{l} - \frac{1}{2}\sin\frac{2\pi x}{l} + \frac{1}{3}\sin\frac{3\pi x}{l} - \cdots\right).$$

We finally remark that Example 1 and Example 3 share the same series, which is not a surprising result because $\phi(x) = x$ is an odd function.

Exercises.

1. Let $\phi(x) = x^2$ for $0 \le x \le 1 = l$.

(a) Calculate its Fourier sine series.

(b) Calculate its Fourier cosine series.

2. Let $\phi(x) = x$ for $0 \le x \le l$.

(a) Calculate its Fourier sine series.

(b) Find the Fourier consine series of the function $x^2/2$, and set $x = 0$ to find the sum of the series $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2}$.

(c) Find the sine series of x^3 and the cosine series of x^4. Then deduce the sum of the series $\sum_{n=1}^{\infty}\frac{(-1)^n}{n^4}$.

3. Find the Fourier cosine series of the function $|\sin x|$ in the interval $(-\pi, \pi)$. Use it to find the sums $\sum_{n=1}^{\infty}\frac{1}{4n^2-1}$ and $\sum_{n=1}^{\infty}\frac{(-1)^n}{4n^2-1}$.

3.2 Separation of Variables, the Dirichlet Condition

In this lecture, we finally come to the physically realistic case of a finite interval $0 < x < l$. The methods we introduce here is the separation of variables.

3.2.1 Wave equation

We first consider the homogeneous Dirichlet conditions for the wave equation:

$$\boxed{\begin{cases} u_{tt} - c^2 u_{xx} = 0 \qquad \text{for } 0 < x < l \\ u(0,t) = 0 = u(l,t) \end{cases}} \tag{3.7}$$

with the initial conditions

$$\boxed{u(x,0) = \phi(x) \qquad u_t(x,0) = \psi(x).} \tag{3.8}$$

Theorem 3.4 *The solution of the boundary value problem is*

$$\boxed{u(x,t) = \sum_n \left(A_n \cos\frac{n\pi ct}{l} + B_n \sin\frac{n\pi ct}{l}\right)\sin\frac{n\pi x}{l}} \tag{3.9}$$

where A_n and B_n satisfy that

$$\phi(x) = \sum_n A_n \sin \frac{n\pi x}{l}, \qquad \psi(x) = \sum_n \frac{n\pi c}{l} B_n \sin \frac{n\pi x}{l}. \tag{3.10}$$

Remark 3.4.1 *These are precisely the Fourier sine series. Therefore,*

$$A_n = \frac{2}{l}\int_0^l \phi(x) \sin \frac{n\pi x}{l} dx \qquad \textit{and} \qquad B_n = \frac{2}{n\pi c}\int_0^l \psi(x) \sin \frac{n\pi x}{l} dx. \tag{3.11}$$

Proof: The method we shall use consists of building up the general solution as a linear combination of special ones that are easy to find. A *separated solution* is a solution of (3.7) and (3.8) of the form

$$u(x,t) = X(x)T(t). \tag{3.12}$$

Our goal is to look for as many separated solutions as possible. Plugging the form (3.12) into the wave equation (3.7), we get

$$X(x)T''(t) = c^2 X''(x)T(t). \tag{3.13}$$

Dividing by $-c^2XT$, we have

$$-\frac{T''}{c^2 T} = -\frac{X''}{X} = \lambda. \tag{3.14}$$

This defines a quantity λ, which must be a *constant*. It is because $\partial\lambda/\partial x = \partial\lambda/\partial t = 0$.

Lemma 3.5 $\lambda > 0$.

Proof: *Case 1.* If $\lambda = 0$, then $X''(x) = 0$. It means that $X = C + Dx$. Notice that $X(0) = X(l) = 0$, so $C = D = 0$. Then $X \equiv 0$. So does $u(x,t)$. It is a contradiction.

Case 2. If $\lambda < 0$, then $\lambda = -\gamma^2$. It means that $X(x) = Ce^{\gamma x} + De^{-\gamma x}$. Notice that $X(0) = X(l) = 0$, so $C = D = 0$. Then $X \equiv 0$. So does $u(x,t)$. It is a contradiction.

Case 3. If λ is a complex number. Let the complex number γ is the root of $-\lambda$. Then $X(x) = Ce^{\gamma x} + De^{-\gamma x}$. Notice that $X(0) = X(l) = 0$, so $C + D = 0 = Cr^{\gamma l} + De^{-\gamma l}$. Therefore $e^{2\gamma l} = 1$. It means $Re(\gamma) = 0$ and $2l \cdot Im(\gamma) = 2\pi n$, where n is an integer. Thus $\lambda = -\gamma^2 = n^2\pi^2/l^2$ is real and positive. It is a contraction

Thus the only case is $\lambda > 0$. ∎

So let $\lambda = \beta^2$, where $\beta > 0$. Then $X(x)$ and $T(t)$ satisfy a pair of *separate* ordinary differential equations:

$$X''(x) + \beta^2 X(x) = 0 \qquad \text{and} \qquad T''(t) + c^2\beta^2 T(t) = 0. \tag{3.15}$$

The solutions have the form

$$X(x) = C\cos\beta x + D\sin\beta x, \tag{3.16}$$

$$T(t) = A\cos\beta ct + B\sin\beta ct, \tag{3.17}$$

where A, B, C, and D are constants. Now consider the boundary conditions $0 = X(0) = C$, and $0 = X(l) = D\sin\beta l$. Thus $\beta l = n\pi$. Then summing all the possible n together, we have

$$u(x,t) = \sum_n \left(A_n \cos\frac{n\pi ct}{l} + B_n \sin\frac{n\pi ct}{l} \right) \sin\frac{n\pi x}{l}.$$

For the initial data, $\phi(x) = u(x,0) = \sum_n A_n \sin\frac{n\pi x}{l}$, and $\psi(x) = u_t(x,0) = \sum_n \frac{n\pi c}{l} B_n \sin\frac{n\pi x}{l}$. It completes the proof of this theorem. ∎

3.2.2 Heat equation

Next, we consider the following boundary value problems of the heat equation

$$\boxed{\begin{cases} u_t - a^2 u_{xx} = 0 \qquad \text{for } 0 < x < l,\ 0 < t < \infty \\ u(0,t) = u(l,t) = 0 \\ u(x,0) = \phi(x) \end{cases}} \tag{3.18}$$

Theorem 3.6 *The solution of the boundary value problem is*

$$\boxed{u(x,t) = \sum_{n=1}^{\infty} A_n e^{-\frac{n^2\pi^2}{l^2}a^2 t} \sin\frac{n\pi x}{l}} \tag{3.19}$$

where A_n satisfies that

$$A_n = \frac{2}{l}\int_0^l \phi(x)\sin\frac{n\pi x}{l}dx. \tag{3.20}$$

Proof: The proof is similar to the previous one. To solve it, we separate the variables $u = T(t)X(x)$ as before. This time we get

$$\frac{T'}{a^2 T} = \frac{X''}{X} = -\lambda = constant.$$

Therefore, $Y(t)$ satisfies the equation $T' = -\lambda a T$, whose solution is $T(t) = Ae^{-\lambda a t}$. Furthermore, $X(x)$ satisfies the equation $-X = \lambda X$ with the same boundary condition $X(0) = X(l) = 0$ as before, and so has the same solutions. ■

3.2.3 Laplace equation

Example 1: The discs

Let us consider the problem

$$\boxed{u_{xx} + u_{yy} = 0 \qquad \text{for } x^2 + y^2 < a^2} \tag{3.21}$$

with the Dirichlet boundary condition $u = h(\theta)$ when $x^2 + y^2 = a^2$.

Our method is to separate variables in polar coordinates: $u = R(r)\Theta(\theta)$. From the equation,

$$0 = u_{xx} + u_{yy} = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = R''\Theta + \frac{1}{r}R'\Theta + \frac{1}{r^2}R\Theta''.$$

Dividing by $R\Theta$ and multiplying by r^2, we find that

$$\Theta'' + \lambda\Theta = 0 \tag{3.22}$$

$$r^2 R'' + rR' - \lambda R = 0. \tag{3.23}$$

For $\Theta(\theta)$ we naturally require periodic boundary conditions, i.e., $\Theta(\theta + 2\pi) = \Theta(\theta)$. Thus

$$\lambda = n^2 \qquad \text{and} \qquad \Theta(\theta) = A\cos n\theta + B\sin n\theta \qquad (n = 1, 2, ...) \tag{3.24}$$

There is also the solution $\lambda = 0$ with $\Theta(\theta) = A$.

Because the equation for R is of the Euler type, the solutions are of the form $R(r) = r^\alpha$. For $\lambda = n^2$, we have

$$\alpha(\alpha-1)r^\alpha + \alpha r^\alpha - n^2 r^\alpha = 0$$

where $\alpha = \pm n$. Thus $R(r) = Cr^n + Dr^{-n}$, and we have the separated solutions

$$u = \left(Cr^n + \frac{D}{r^n}\right) \cdot (A\cos n\theta + B\sin n\theta)$$

for $n = 1, 2, 3, \ldots$. For the case $n = 0$, it is easy to see that the solution is

$$u = C + D\log r.$$

If we only consider the solutions that are finite at the origin ($r = 0$), summing the remaining soltuions, we have

$$u = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} r^n(A_n\cos n\theta + B_n\sin n\theta). \tag{3.25}$$

where the coefficients A_n satisfy the boundary condition on the circle $r = a$ that

$$h(\theta) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} a^n(A_n\cos n\theta + B_n\sin n\theta).$$

This is precisely the full Fourier series for $h(\theta)$, so we know that

$$\boxed{A_n = \frac{1}{\pi a^n}\int_{-\pi}^{\pi} h(\theta)\cos n\theta d\theta, \qquad B_n = \frac{1}{\pi a^n}\int_{-\pi}^{\pi} h(\theta)\sin n\theta d\theta.} \tag{3.26}$$

Now comes an amazing fact. The series can be summed explicitly! In fact,

$$\begin{aligned} u(r,\theta) &= \int_{-\pi}^{\pi} h(\phi)\frac{d\phi}{2\pi} + \sum_{n=1}^{\infty}\frac{r^n}{\pi a^n}\int_{-\pi}^{\pi} h(\phi)(\cos n\phi\cos n\theta + \sin n\phi\sin n\theta)d\phi \\ &= \int_{-\pi}^{\pi} h(\phi)\left\{1 + 2\sum_{n=1}^{\infty}\left(\frac{r}{a}\right)^n\cos n(\theta-\phi)\right\}\frac{d\phi}{2\pi}. \end{aligned}$$

Notice that

$$\begin{aligned} 1 + 2\sum_{n=1}^{\infty}\left(\frac{r}{a}\right)^n\cos n(\theta-\phi) &= 1 + \sum_{n=1}^{\infty}\left(\frac{r}{a}e^{i(\theta-\phi)}\right)^n + \sum_{n=1}^{\infty}\left(\frac{r}{a}e^{-i(\theta-\phi)}\right)^n \\ &= 1 + \frac{re^{i(\theta-\phi)}}{a - re^{i(\theta-\phi)}} + \frac{re^{-i(\theta-\phi)}}{a - re^{-i(\theta-\phi)}} \\ &= \frac{a^2 - r^2}{a^2 - 2ar\cos(\theta-\phi) + r^2}. \end{aligned}$$

Therefore,

$$\boxed{u(r,\theta) = (a^2 - r^2)\int_{-\pi}^{\pi}\frac{h(\phi)}{a^2 - 2ar\cos(\theta-\phi) + r^2}\frac{d\phi}{2\pi}.} \tag{3.27}$$

This single formula is known as *Poisson's formula*.

Finally, we would like to remark that the formula derived here is equivalent to the one in Lecture 3 by the Green's method. Let $\mathbf{x}$ be the coordinate of (r, θ), while $\mathbf{x}'$ is the polar coordinate of (a, ϕ) on the boundary. Notice that $|\mathbf{x} - \mathbf{x}'| = a^2 + r^2 - 2ar\cos(\theta - \phi)$. Thus we have

$$\boxed{u(\mathbf{x}) = \frac{a^2 - |\mathbf{x}|^2}{2\pi a} \int_{|\mathbf{x}'|=a} \frac{u(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|^2} dS.} \tag{3.28}$$

Example 2: The annulus

The Dirichlet problem for an annulus is

$$\begin{aligned} u_{xx} + u_{yy} &= 0 \qquad \text{in } 0 < a^2 < x^2 + y^2 < b^2 \\ u &= g(\theta) \qquad \text{for } x^2 + y^2 = a^2 \\ u &= h(\theta) \qquad \text{for } x^2 + y^2 = b^2. \end{aligned}$$

The separated solutions are just the same as for a circle except that we don't throw out the functions r^{-n} and $\log r$, as these functions are perfectly finite within the annulus. So the solution is

$$\boxed{u(r, \theta) = \frac{1}{2}(C_0 + D_0 \log r) + \sum_{n=1}^{\infty} (C_n r^n + D_n r^{-n}) \cos n\theta + (A_n r^n + B_n r^{-n}) \sin n\theta.} \tag{3.29}$$

The coefficients are determined by setting $r = a$ and $r = b$.

Exercises.

1. Solve the problem $u_{tt} - c^2 u_{xx} = 0$, with the boundary conditions $u(0, t) = u(l, t) = 0$ and the initial conditions $u(x, 0) = x$ and $u_t(x, 0) = 0$.

2. Consider waves in a resistant medium that satisfy the problem $u_{tt} - c^2 u_{xx} + r u_t = 0$, $u(0, t) = u(l, t) = 0$, $u(x, 0) = \phi(x)$, and $u_t(x, 0) = \psi(x)$, where r is a constant.

(a) If $0 < r < 2\pi c/l$, write down the series expansion of the solution.

(b) If $2\pi c/l < r < 4\pi c/l$, write down the series expansion of the solution.

3. Separate variables for the equation $tu_t = u_{xx} + 2u$ with the boundary condition $u(0, t) = u(\pi, t) = 0$. Show that there are an infinite number of solutions that satisfy the initial condition $u(x, 0) = 0$.

4. Solve $u_{xx} + u_{yy} = 0$ in the disk $\{r < a\}$ with the boundary condition $u = 1 + 3\sin\theta$ on $r = a$.

5. Determine the coefficients in the annulus problem of the lecture note.

3.3 The Neumann Boundary Conditions

3.3.1 Heat equation and wave equation.

we consider the following boundary value problems of the heat equation

$$\boxed{\begin{cases} u_t - a^2 u_{xx} = 0 \qquad \text{for } 0 < x < l,\ 0 < t < \infty \\ u_x(0, t) = u_x(l, t) = 0 \\ u(x, 0) = \phi(x) \end{cases}} \tag{3.30}$$

Theorem 3.7 *The solution of the boundary value problem is*

$$\boxed{u(x,t) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n e^{-\frac{n^2\pi^2a^2}{l^2}t} \cos\frac{n\pi x}{l}} \tag{3.31}$$

where A_n satisfies that

$$A_n = \frac{2}{l}\int_0^l \phi(x)\cos\frac{n\pi x}{l}dx. \tag{3.32}$$

Proof: As before, let us consider the solutions of the form $u(x,t) = X(x)T(t)$. Then now $X(x)$ satisfies the following problem:

$$\boxed{-X'' = \lambda X, \qquad X'(0) = X'(l) = 0.} \tag{3.33}$$

First, if $\lambda < 0$ or if λ is complex, then it can be shown directly, as in the Dirichlet case, that there is no eigenfunction.

Next if $\lambda = 0$, then $X''(x) = 0$. So $X(x) = C + Dx$ with the Neumann boundary conditions $X'(x) = D = 0$ on $x = 0$ and $x = l$. Therefore $\lambda = 0$ is an eigenvalue, and any constant function is its eigenfunction.

If $\lambda = \beta^2 > 0$, then $X(x) = C\cos\beta x + D\sin\beta x$, so that

$$X'(x) = -C\beta\sin\beta x + D\beta\cos\beta x.$$

From the boundary conditions, $0 = X'(0) = D\beta$, so $D = 0$, and at the same time, $0 = X'(l) = -C\beta\sin\beta l$. Since we don't want $C = 0$, we must have $\sin\beta l = 0$. Thus $\beta = n\pi/l$, where $n = 1, 2, 3, ...$

Therefore, we have the eigenvalue $\lambda_n = (\frac{n\pi}{l})^2$, for $n = 0, 1, 2, 3, ...$, and summing all the possible solutions together, the the solutions of the heat equation with the Neumann boundary conditions are as said in the theorem. While, the solution requires the initial data to have the 'Fourier cosine expansion'

$$\phi(x) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \cos\frac{n\pi x}{l}. \tag{3.34}$$

It completes the proof of this theorem. ■

Next following the same argument, we can have a similar results to the wave equation with the Neumann boundary conditions.

Theorem 3.8 *The solution of the boundary value problem*

$$\boxed{\begin{cases} u_{tt} - c^2u_{xx} = 0 \qquad \textit{for } \ 0 < x < l, \ t > 0 \\ u_x(0,t) = 0 = u_x(l,t) \\ u(x,0) = \phi(x) \qquad u_t(x,0) = \psi(x) \end{cases}} \tag{3.35}$$

is

$$\boxed{u(x,t) = \frac{1}{2}A_0 + \frac{1}{2}B_0t + \sum_{n=1}^{\infty}\left(A_n\cos\frac{n\pi ct}{l} + B_n\sin\frac{n\pi ct}{l}\right)\cos\frac{n\pi x}{l}} \tag{3.36}$$

where A_n and B_n satisfy that

$$A_n = \frac{2}{l}\int_0^l \phi(x)\cos\frac{n\pi x}{l}dx \qquad \textit{and} \qquad B_n = \frac{2}{n\pi c}\int_0^l \psi(x)\cos\frac{n\pi x}{l}dx, \tag{3.37}$$

while $B_0 = \frac{2}{l}\int_0^l \psi(x)dx$.

3.3.2 Laplace equation

Example 1: The wedge

Consider the Laplace equation in a wedge: $\{0 < \theta < \beta,\ 0 < r < a\}$, with the homogenous Dirichlet condition on the straight sides and the inhomogeneous Neumann condition on the curved side. That is,

$$\boxed{u(r,0) = 0 = u(r,\beta), \qquad \frac{\partial u}{\partial r}(a,\theta) = h(\theta).} \tag{3.38}$$

The separation of variable technique works just as for the circle, namely,

$$\Theta'' + \lambda\Theta = 0, \qquad r^2R'' + rR' - \lambda R = 0,$$

where Θ satisfies the boundary conditions $\Theta(0) = \Theta(\beta) = 0$. This is our standard eigenvalue problem, which has the solutions

$$\lambda = (\frac{n\pi}{\beta})^2, \qquad \Theta(\theta) = \sin\frac{n\pi\theta}{\beta}. \tag{3.39}$$

Next the radial equation $r^2R'' + rR' - \lambda R = 0$ admits the solutions $R(r) = r^\alpha$, where $\alpha^2 - \lambda = 0$. The negative exponent is rejected again because we are looking for a solution $u(r,\theta)$ that is continuous in the wedge as well as its boundary (the function $e^{-n/\beta}$ is infinite at the origin). Thus we end up with the series

$$\boxed{u(r,\theta) = \sum_{n=1}^{\infty} A_n r^{n\pi/\beta} \sin\frac{n\pi\theta}{\beta}.} \tag{3.40}$$

Finally, the inhomogeneous boundary condition requires that

$$h(\theta) = \sum_{n=1}^{\infty} A_n \frac{n\pi}{\beta} a^{-1+n\pi/\beta} \sin\frac{n\pi\theta}{\beta}.$$

This is just a Fourier sine series in the interval $[0,\beta]$, so its coefficients are given by the formula

$$\boxed{A_n = a^{1-n\pi/\beta}\frac{2}{n\pi}\int_0^\beta h(\theta)\sin\frac{n\pi\theta}{\beta}d\theta.} \tag{3.41}$$

Exercises.

1. Solve $u_{tt} - c^2u_{xx} = 0$ with the boundary conditions $u_x(0,t) = u_x(\pi,t) = 0$ and the initial conditions $u(x,0) = 0$, $u_t(x,0) = \cos^2 x$.

2. Solve the diffusion problem $u_t = a^2u_{xx}$ in $0 < x < l$, with the mixed boundary conditins $u(0,t) = u_x(l,t) = 0$.

3. Consider the heat equation $u_t - a^2u_{xx} = 0$ with the periodic boundary conditions $u(-l,t) = u(l,t)$ and $u_x(-l,t) = u_x(l,t)$.

(a) Show that the eigenvalues are $\lambda = (\frac{n\pi}{l})^2$, for $n = 0,1,2,3,....$

(b) Show that the solution is

$$u(x,t) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}\left(a_n\cos\frac{n\pi x}{l} + b_n\sin\frac{n\pi x}{l}\right)e^{-n^2\pi^2a^2t/l^2}.$$

4. Solve $u_{xx} + u_{yy} = 0$ in the disk $r < a$ with the boundary condition $\frac{\partial u}{\partial r} - hu = f(\theta)$, where $f(\theta)$ is an arbitrary smooth function.

5. Solve $u_{xx} + u_{yy} = 0$ in the wedge $r < a$, $0 < \theta < \beta$ with the boundary conditions $u = \theta$ on $r = a$, $u = 0$ on $\theta = 0$, and $u = \beta$ on $\theta = \beta$.

3.4 Inhomogeneous Boundary Problems, Expansion Method

In this section, we consider problems with sources given at the boundary. It is easy to see that naive use of the separation of variables technique will not work. So we try a slightly different approach.

3.4.1 Heat equation

Let's begin with the diffusion equation with sources at both endpoints.

$$\boxed{\begin{cases} u_t - a^2 u_{xx} = 0 \qquad \text{for } \ 0 < x < l, \ t > 0 \\ u(0,t) = h(t) \qquad u(l,t) = j(t) \\ u(x,0) = 0 \end{cases}} \tag{3.42}$$

We already know that for the corresponding homogeneous problem the correct expansion is the Fourier sine series. For each t, we certainly can expand

$$u(x,t) = \sum_{n=1}^{\infty} u_n(t) \sin \frac{n\pi x}{l}, \tag{3.43}$$

for some coefficients $u_n(t)$, where $u_n(t) = \frac{2}{l}\int_0^l u(x,t) \sin \frac{n\pi x}{l} dx$ and $u(0) = 0$. Next let's expand u_t too. Thus

$$u_t = \sum_{n=1}^{\infty} v_n(t) \sin \frac{n\pi x}{l} dx \tag{3.44}$$

where $v_n(t) = \frac{2}{l}\int_0^l u_t \sin \frac{n\pi x}{l} dx = \frac{du_n}{dt}$. Hence by the heat equation and integrating by part twice,

$$\frac{du_n}{dt} = \frac{2a^2}{l}\int_0^l u_{xx} \sin \frac{n\pi x}{l} dx = \frac{2a^2}{l}\left(u_x \sin \frac{n\pi l}{l} - \frac{n\pi}{l} u \cos \frac{n\pi x}{l} \right)\Big|_0^l - \frac{2a^2}{l}\int_0^l (\frac{n\pi}{l})^2 u(x,t) \sin \frac{n\pi x}{l} dx.$$

Hence $u_n(t)$ satisfies the ODE

$$\frac{du_n}{dt} + (\frac{an\pi}{l})^2 u_n = -\frac{2a^2 n\pi}{l^2}(-1)^n j(t) + \frac{2a^2 n\pi}{l^2} h(t),$$

with the initial data $u_n(0) = 0$. Its solution is

$$u_n(t) = -\frac{2a^2 n\pi}{l^2}\int_0^t e^{-(\frac{an\pi}{l})^2 (t-s)} [(-1)^n j(s) - h(s)] ds.$$

Meanwhile, the solution of the heat equation is $u(x,t) = \sum_{n=1}^{\infty} u_n(t) \sin \frac{n\pi x}{l}$.

3.4.2 Wave equation

As a second case, let's solve the inhomogeneous wave problem

$$\boxed{\begin{cases} u_{tt} - c^2 u_{xx} = f(x,t) \qquad \text{for } \ 0 < x < l, \ t > 0 \\ u(0,t) = h(t) \qquad u(l,t) = k(t) \\ u(x,0) = \phi(x) \qquad u_t(x,0) = \psi(x) \end{cases}} \tag{3.45}$$

Again we expect that we can expand the solution:

$$u(x,t) = \sum_{n=1}^{\infty} u_n(t) \sin \frac{n\pi x}{l}, \tag{3.46}$$

for some coefficients $u_n(t)$, where $u_n(t) = \frac{2}{l}\int_0^l u(x,t) \sin \frac{n\pi x}{l} dx$. Then by the equation,

$$\frac{d^2 u_n}{dt^2} = \frac{2}{l}\int_0^l u_{tt} \sin \frac{n\pi x}{l} dx = \frac{2c^2}{l}\int_0^l u_{xx} \sin \frac{n\pi x}{l} dx + \frac{2}{l}\int_0^l f \sin \frac{n\pi x}{l} dx.$$

Let $f_n(t) := \frac{2}{l}\int_0^l f(x,t) \sin \frac{n\pi x}{l} dx$. By the integrating by part and the boundary conditions of u on $x = 0$ and $x = l$ as before, u_n satisfies the following ODE:

$$\frac{d^2 u_n}{dt^2} + (\frac{cn\pi}{l})^2 u_n = -\frac{2c^2 n\pi}{l^2}[(-1)^n k(t) + h(t)] + f_n(t),$$

with the initial conditions

$$u_n(0) = \phi_n \qquad u_n'(0) = \psi_n,$$

where $\phi_n(t) = \frac{2}{l}\int_0^l \phi \sin \frac{n\pi x}{l} dx$ and $\psi_n(t) = \frac{2}{l}\int_0^l \psi \sin \frac{n\pi x}{l} dx$.

Exercises.

1. Solve the equation $u_t - u_{xx} = 0$ with $u_x(0,t) = 0$, $u(1,t) = 1$, and $u(x,0) = x^2$. Compute the first two coefficients explicityly. Find the equilibrium state (the term that does not tend to zero when t goes to the infinity).

2. Solve $u_{tt} - c^2 u_{xx} = k$ for $0 < x < l$, with the boundary conditions $u(0,t) = u_x(l,t) = 0$ and the initial conditions $u(x,0) = 0$, $u_t(x,0) = V$. Here k and V are constants.

3. Solve $u_{tt} - c^2 u_{xx} = g(x) \sin wt$, with $u(0,t) = u(l,t) = 0$ at both ends, and $u = u_t = 0$ when $t = 0$. For which values of w can resonance occur? (Resonance means growth in time.)

4. Carry out the solution of the inhomogeneous wave problem explicitely in the case that $f(x,t) = F(x) \cos wt$, $h(t) = H \cos wt$, and $k(t) = K \cos wt$.